2408989 : [illegible]

DATE

PAGE No 1

Tutorial - 4

(1) Forward pass (compute hidden layers)

Given,

$i_1 = 0.1$

$i_2 = 0.5$

Weights from input layer to hidden layer:

$w_1 = 0.1$

$w_2 = 0.2$

$w_3 = 0.3$

$w_4 = 0.4$

Bias for hidden layer:

$b_1 = 0.25$

Activation function:

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

(1) Net input for hidden neuron h_1

$zh_1 = (i_1 \cdot w_1) + (i_2 \cdot w_3) + b_1$

substitutes values:

$$zh_1 = (0.1 \cdot 0.1) + (0.5 \cdot 0.3) + 0.25$$

$$zh_1 = 0.01 + 0.15 + 0.25$$

$$= 0.41$$

Now apply sigmoid:

$$h_1 = \sigma(0.41) = \frac{1}{1+e^{-0.41}}$$

$$h_1 = 0.6011$$

2408989: DEHAMWANENG

(2) Compute net input for hidden neuron h_2

$zh_2 = (i_1 \cdot w_2) + (i_2 \cdot w_4) + b_1$

substitute values,

$zh_2 = (0.1 \cdot 0.2) + (0.5 \cdot 0.4) + 0.25$

$zh_2 = 0.02 + 0.20 + 0.25$

Now apply sigmoid:

$h_2 = \sigma(0.47) = \frac{1}{1+e^{-0.47}}$

$h_2 = 0.6154$

hidden layer outputs

$h_1 = 0.6011$

$h_2 = 0.6154$

forward pass ; Hidden layer outputs

$zh_1 = (0.1 \times 0.1) + (0.5 \times 0.3) + 0.25$

$= 0.41$

$h_1 = \sigma(0.41) = \frac{1}{1+e^{-0.41}}$

$= 0.6011$

$zh_2 = (0.1 \times 0.2) + (0.5 \times 0.4) + 0.25$

$= 0.47$

$h_2 = \sigma(0.47)$

$= \frac{1}{1+e^{-0.47}} = 0.6154$

∴ Hidden layer outputs = [0.6011, 0.6154]

2408980: WEMIN WAHEM

② Forward pass (final output)

Given,

From step 1,

$h_1 = 0.6011$, $h_2 = 0.6154$

Weights from hidden layer to output layer:

$w5 = 0.5$

$w6 = 0.6$

$w7 = 0.7$

$w8 = 0.8$

Bias for output layer:

$b2 = 0.35$

Activation function:

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

① Compute net input for output neuron o_1.

$$z_{o1} = (h_1 \cdot w_5) + (h_2 \cdot w_7) + b_2$$

Substitute values:

$$z_{o1} = (0.6011 \cdot 0.5) + (0.6154 \cdot 0.7) + 0.5$$

$$= 0.3006 + 0.4308 + 0.35$$

$$= 1.0814$$

Now apply sigmoid:

$$o_1 = \sigma(1.0814) = \frac{1}{1+e^{-1.0814}}$$

$$o_1 = 0.7462$$

2408989 : DEHAN WANEM

(2) Compute net input for output neuron o_2

$$z_{o2} = (h_1 \cdot w_7) + (h_2 \cdot w_8) + b_2$$

substitu values:

$$z_{o2} = (0.6011 \cdot 0.6) + 0.6154 \cdot 0.8) + 0.35$$

$$z_{o2} = 0.3607 + 0.4923 + 0.35$$

$$= 1.2030$$

Now apply sigmoid:

$$o_2 = \sigma(1.2030) = \frac{1}{1+e^{-1.2030}}$$

$$o_2 = 0.7691$$

final outputs, $o_1 = 0.7468$

$o_2 = 0.7691$

Forward pass: Final output,

$$z_{o1} = (0.6011 \times 0.5) + (0.6154 \times 0.7) + 0.35$$

$$z_{o1} = 0.3006 + 0.4308 + 0.35$$

$$= 1.0814$$

$$o_1 = \sigma(1.0814) = \frac{1}{1+e^{-1.0814}} = 0.7468$$

$$z_{o2} = (0.6011 \times 0.6) + (0.6154 \times 0.8) + 0.35$$

$$z_{o2} = 0.3607 + 0.4923 + 0.35$$

$$= 1.2030$$

$$o_2 = \sigma(1.2030) = \frac{1}{1+e^{-1.2030}}$$

$$= 0.7691$$

Final output = [0.7468, 0.7691]

24089889 : DEHAN WANEMA

DATE / /
PAGE No. 5

Loss calculation,

Given,

Target outputs:

$t_1 = 0.05$, $t_2 = 0.95$

predicted outputs from forward pass:

$o_1 = 0.7468$, $o_2 = 0.7691$ //

$$E_{total} = \frac{1}{2}(t_1 - o_1)^2 + \frac{1}{2}(t_2 - o_2)^2$$ //

(1) Error for output neuron 1

$$E_1 = \frac{1}{2}(t_1 - o_1)^2$$

Substitue values:

$$E_1 = \frac{1}{2}(t_1 - o_1)^2$$

$$E_1 = \frac{1}{2}(0.05 - 0.7468)^2$$

$$E_1 = \frac{1}{2}(-0.6968)^2$$

$$E_1 = \frac{1}{2}(0.4855)$$

$$E_1 = 0.2428$$ //

(2) Error for output neuron 2

$$E_2 = \frac{1}{2}(t_2 - o_2)^3$$

substitute values:

$$E_2 = \frac{1}{2}(0.95 - 0.7691)^2$$

$$E_2 = \frac{1}{2}(0.1809)^2$$

$E_2 = \frac{1}{2}(0.0327)$, $E_2 = 0.0164$ //

2408989 ! DEIHAN WANEM

Total error:

$E_{total} = E_1 + E_2$

$E_{total} = 0.2428 + 0.0164$

$E_{total} = 0.2592$

Loss calculation,

$E_1 = \frac{1}{2}(0.05 - 0.7468)^2$

$= 0.2428$ //

$E_2 = \frac{1}{2}(0.95 - 0.7691)^2$

$= 0.0164$

$E_{total} = E_1 + E_2$

$E_{total} = 0.2428 + 0.0164$

$= 0.2592$

$\therefore$ Total error $= 0.2592$ //

2908989 : DEHAN WANEM

(3) Backpropagation for output layer weight:

Now compute gradient for:

w_5, w_6, w_7, w_8

we use:

$$\frac{\partial E}{\partial w} = \frac{\partial E}{\partial o} \cdot \frac{\partial o}{\partial o} \cdot \frac{\partial z}{\partial w}$$

Given,

from previous steps:

$$o_1 = 0.7468, \quad o_2 = 0.7691$$

$$t_1 = 0.05, \quad t_2 = 0.95$$

$$h_1 = 0.6011, \quad h_2 = 0.6154$$

Also:

$o(1-o)$.

for sigmoid derivative,

Gradient for w_5,

since w_5 connects h_1 to output o_1,

$$\frac{\partial E}{\partial w_5} = (o_1 - t_1)\, o_1 (1 - o_1)\, h_2$$

substitue values:

$$\frac{\partial E}{\partial w_5} = (0.7468 - 0.05)(0.7468)(1 - 0.7468)(0.6011)$$

$$= (0.6968)(0.7468)(0.2532)(0.6011)$$

$$\frac{\partial E}{\partial w_2} = 0.0780$$

2408989! DEHAN WANEM

DATE	/ /
PAGE No.	8

Gradient for ω_6,

since w_6 connect h_2 to output o_1,

$$\frac{\partial E}{\partial w_6} = (o_1 - t_1)\, o_1(1 - o_1)\, h_2$$

$$\frac{\partial E}{\partial w_6} = (0.7468 - 0.05)(0.7468)(1 - 0.7468)(0.6154)$$

$$= (0.6968)(0.7468)(0.2532)(0.6154)$$

$$= \frac{\partial E}{\partial w_6} = 0.0798$$

Gradient for ω_7,

since, ω_7 connects h_1 to output o_2,

$$\frac{\partial E}{\partial w_7} = (o_2 - t_2)\, o_2(1 - o_2)\, h_1$$

substitute values!

$$\partial E = (0.7691 - 0.95)(0.7691)(1 - 0.7691)(0.6011)$$

$$\frac{\partial E}{\partial w_7} = -0.0193$$

Gradient for ω_8,

since ω_8 connect h_2 to output o_2,

$$\frac{\partial E}{\partial w_8} = (o_2 - t_2)\, o_2(1 - o_2)\, h_2$$

$$\frac{\partial E}{\partial w_8} = (0.7692 - 0.95)(0.7691)(1 - 0.7691)(0.6154)$$

$$= (-0.1809)(0.7691)(0.2309)(0.6154)$$

2408089 DEHAN WANEM

$\frac{\partial E}{\partial w_8} = -0.0198$ //

Backpropagation for output-layer weights.

$\frac{\partial E}{\partial w_5} = (o_1 - t_1)\ o_1(1-o_1)\ h_1$

$\frac{\partial E}{\partial w_5} = (0.7468 - 0.05)(0.7468)(1-0.7468)(0.6011)$

$= 0.0780$

$\frac{\partial E}{\partial w_6} = (o_1 - t_1)\ o_1(1-o_1)\ h_2$

$\frac{\partial E}{\partial w_6} = (0.7468 - 0.05)(0.7468)(1-0.7468)(0.6154)$

$= 0.0798$

$\frac{\partial E}{\partial w_7} = (o_2 - t_2)\ o_2(1-o_2)\ h_1$

$\frac{\partial E}{\partial w_7} = (0.7691 - 0.95)(0.7691)(1-0.7691)(0.6011)$

$= -0.0193$

$\frac{\partial E}{\partial w_8} = (o_2 - t_2)\ o_2(1-o_2)\ h_2$

$\frac{\partial E}{\partial w_8} = (0.7691 - 0.95)(0.7691)(1-0.7691)(0.6154)$

$= -0.0198$

$\therefore \frac{\partial E}{\partial w_5} = 0.0780,\ \frac{\partial E}{\partial w_6} = 0.0798,\ \frac{\partial E}{\partial w_7} = -0.0193,\ \frac{\partial E}{\partial w_8} = -0.0198$

24080821 DEHAN WANEM

DATE / /
PAGE No

Weight update for w_5, w_6, w_7, w_8,
use gradient descent,

$$w_{new} = w_{old} - \eta \frac{\partial E}{\partial w}$$

Assume learning rate = $\eta = 0.5$
initial weights:

$$w_5 = 0.5,\ w_6 = 0.6,\ w_7 = 0.7,\ w_8 = 0.8$$

Using the gradients:

Update weights,

$w_5^{new} = 0.5 - (0.5 \times 0.0780) = 0.4610$

$w_6^{new} = 0.6 - (0.5 \times 0.0798) = 0.5601$

$w_7^{new} = 0.7 - (0.5 \times (-0.0193)) = 0.7097$

$w_8^{new} = 0.8 - (0.5 \times (-0.0198)) = 0.8099$

~~updating output-layer weights:~~
~~$w_5^{new} = 0.5 - (0.5 \times 0.0780) = 0.$~~

Backpropagation for first-layer weights
(w_1, w_2, w_3, w_4)

$\delta o_1 = (o_1 - t_1)\, o_1 (1 - o_1),\ \delta o_2 = (o_2 - t_2)\, o_2 (1 - o_2)$

$\delta h_1 = (\delta o_1 w_5 + \delta o_2 w_7)\, h_1 (1 - h_1)$

$\delta h_2 = (\delta o_1 w_6 + \delta o_2 w_8)\, h_2 (1 - h_2)$

2408989 : DEHAN WANEM

substituting values:

$\delta_{o1} = 0.1310$, $\delta_{o2} = -0.0321$

$\delta h_1 = [0.1310 \times 0.5 + (-0.0321) \times 0.7] \times 0.6011 (1 - 0.6011)$
$= 0.0103$

$\delta h_2 = [0.1310 \times 0.6 + (-0.0321) \times 0.8] \times 0.6154 (1 - 0.6154)$
$= 0.0124$

$\frac{\partial e}{\partial w_1} = 0.0103 \times 0.1 = 0.0010$

$\frac{\partial e}{\partial w_2} = 0.0103 \times 0.5 = 0.0052$

$\frac{\partial e}{\partial w_3} = 0.0124 \times 0.1 = 0.0012$

$\frac{\partial e}{\partial w_4} = 0.0124 \times 0.5 = 0.0062$

$\therefore \frac{\partial e}{\partial w_1} = 0.0010$, $\frac{\partial e}{\partial w_2} = 0.0052$, $\frac{\partial e}{\partial w_3} = 0.0012$, $\frac{\partial e}{\partial w_4} = 0.0062$

② Update first-layer weights (w_1, w_2, w_3, w_4)

using,

$$w_{new} = w_{old} - \eta \frac{\partial e}{\partial w}$$

with learning rate

$$\eta = 0.5$$

Given:

$w_1 = 0.1$, $w_2 = 0.2$, $w_3 = 0.3$, $w_4 = 0.4$

2008987: DEHAN WANEM

So,

$w_1^{new} = 0.1 - (0.5 \times 0.0010) = 0.0995$

$w_2^{new} = 0.2 - (0.5 \times 0.0052) = 0.1974$

$w_3^{new} = 0.3 - (0.5 \times 0.0012) = 0.2994$

$w_4^{new} = 0.4 - (0.5 \times 0.0062) = 0.3969$

$\therefore w_1^{new} = 0.0995$, $w_2^{new} = 0.1974$,

$w_3^{new} = 0.2994$, $w_4^{new} = 0.3969$

(3) Final updated weights after one epoch,

updated first-layer weights,

$w_1 = 0.0995$, $w_2 = 0.1974$, $w_3 = 0.2994$,
$w_4 = 0.3969$

updated output-layer weights,

$w_5 = 0.4610$, $w_6 = 0.5601$, $w_7 = 0.7097$
$w_8 = 0.8099$

$\therefore$ final updated weight after one epoch are:

$[w_1, w_2, w_3, w_4, w_5, w_6, w_7, w_8]$

$= [0.0995, 0.1974, 0.2994, 0.3969,$
$0.4610, 0.5601, 0.7097, 0.8099]$